

Surgical factors play a critical role in predicting functional outcomes using machine learning in robotic-assisted total knee arthroplasty

Lisa Spahn Lundgren¹ | Nathalie Willems¹ | Robert C. Marchand² |
Cécile Batailler^{3,4}  | Sébastien Lustig^{3,4}

¹Stryker Department, Amsterdam, The Netherlands

²Orthopedic Surgery Department, South County Orthopaedics, Ortho Rhode Island, Wakefield, Rhode Island, USA

³Orthopedic Surgery Department, Croix-Rousse Hospital, Lyon, France

⁴Univ Lyon, IFSTTAR, LBMC UMR_T9406, Université Claude Bernard Lyon 1, Villeurbanne, France

Correspondence

Cécile Batailler, Univ Lyon, Université Claude Bernard Lyon 1, IFSTTAR, LBMC UMR_T9406, 69622, Villeurbanne, France.
Email: cecile-batailler@hotmail.fr

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None

Abstract

Purpose: Predictive models help determine predictive factors necessary to improve functional outcomes after total knee arthroplasty (TKA). However, no study has assessed predictive models for functional outcomes after TKA based on the new concepts of personalised surgery and new technologies. This study aimed to develop and evaluate predictive modelling approaches to predict the achievement of minimal clinically important difference (MCID) in patient-reported outcome measures (PROMs) 1 year after TKA.

Methods: Four hundred thirty robotic-assisted TKAs were analysed in this retrospective study. The mean age was 67.9 ± 7.9 years; the mean body mass index (BMI) was $32.0 \pm 6.8 \text{ kg/m}^2$. The following PROMs were collected preoperatively and 1-year postoperatively: knee injury and osteoarthritis outcome score for joint replacement, Western Ontario and McMaster Universities osteoarthritis index (WOMAC) Function, WOMAC Pain. Demographic data, preoperative CT scan, implant size, implant position on the robotic system and characteristics of the joint replacement procedure were selected as predictive variables. Four machine learning algorithms were trained to predict the MCID status at 1-year post-TKA for each PROM survey. 'No MCID' was chosen as the target. Models were evaluated by class discrimination (F1-score) and area under the receiver operating characteristic curve (ROC-AUC).

Results: The best-performing model was ridge logistic regression for WOMAC Function (area under the curve [AUC] = 0.80, F1 = 0.48, sensitivity = 0.79, specificity = 0.62). Variables most strongly contributing to not achieving MCID status were preoperative PROMs, high BMI and femoral resection depth (posterior and distal), supporting functional positioning principles. Conversely, variables contributing to a positive outcome (achieving MCID) were medial/lateral alignment of the tibial component, whether the procedure was an outpatient surgery and whether the patient received managed Medicare insurance.

Conclusion: The most predictive variables included preoperative PROMs, BMI and surgical planning. The surgical predictive variables were valgus

Abbreviations: BMI, body mass index; CR, cruciate retaining; HKA, hip-knee-ankle angle; JLCA, joint line congruency angle; JLO, joint line obliquity; JSW, joint space widths; LDFA, lateral distal femur angle; MCID, minimal clinically important difference; MPTA, medial proximal tibial angle; PROMs, patient-reported outcome measures; PS, posterior stabilised; RA-TKA, robotic-assisted total knee arthroplasty; TKA, total knee arthroplasty.

Institution(s) at which the work was performed: Orthopedic Surgery Department, South County Hospital, 100 Kenyon Ave, WAKEFIELD, RI 02879, USA.

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femoral alignment and femoral rotation, reflecting the benefits of personalised surgery. Including surgical variables in predictive models for functional outcomes after TKA should guide clinical and surgical decision-making for every patient.

Level of Evidence: Level III.

KEYWORDS

minimal clinically important difference, personalised surgery, predictive model, robotic-assisted surgery, total knee arthroplasty

INTRODUCTION

Total knee arthroplasty (TKA) leads to good results in returning to daily activities and overall reasonable functional improvement. Despite advances in TKA, surgical technique and postoperative management over the years, up to 20% of TKA patients remain dissatisfied after TKA [15]. In a multicentre series of 347 TKAs, only 62% of the patients were pain free during gait and 35% were pain free while climbing stairs [8]. Only 48% of the patients declared being 'very satisfied' with the procedure. Therefore, although TKA is already an effective surgery, the aim is now to improve patient satisfaction and functional outcomes.

New technologies struggle to improve functional outcomes [6]. Indeed, even if the implant's three-dimensional (3D) positioning is more accurate and reproducible, functional outcomes and satisfaction are not systematically enhanced [3, 5, 6]. Personalised surgery is probably not only personalised two-dimensional (2D) alignment but also personalised implant 3D positioning and personalised perioperative management of the patient. Many factors impact functional outcomes. Artificial intelligence with machine learning and predictive models can be insightful, helping determine which factors are predictive of functional outcomes [38, 41]. Predictive modelling aims to generate estimates for a particular target output [18]. These models learn to identify the relationships between a set of variables (e.g., age, body mass index [BMI], etc.) and the selected target (e.g., postoperative satisfaction) [7]. Hospital teams can proactively leverage such decision-aid tools to manage patients (e.g., Is the surgery indicated? Which surgical planning is the most appropriate for this patient?) and their postoperative expectations. Several studies have published predictive models for TKA outcomes that have considered variables such as functional scores, demographic characteristics and psychological variables. However, no study has developed a predictive model to predict the improvement of the functional outcomes after TKA based on current patient-reported outcome measures (PROMs), new concepts of personalised surgery and new technologies [4].

The aims of this study were (1) to develop and evaluate predictive modeling approaches to predict the achievement of minimal clinically important difference (MCID) in PROMs 1 year after robotic-assisted TKA (RA-TKA) and (2) to identify the main predictive factors that predict the failure to reach MCID in PROMs at 1 year after RA-TKA.

METHODS

Population

The database from 2016 to 2020 was queried for patients who had undergone unilateral primary RA-TKA and for which functional and general health PROMs have been collected preoperatively and postoperatively. The follow-up time point was chosen as 1-year post-TKA. Patients missing preoperative or 1-year postoperative up PROMs were excluded ($n=20$; 4.5%). Patients with high BMI (≥ 40) or age (≥ 89) were masked in the database and therefore excluded from this analysis. Excluding patients with missing data fields from an initial cohort of 443 patients, the final patient cohort included 423 robotic-assisted primary TKAs from two experienced and high-volume surgeons, with the same experience and surgical technique (Table 1). The patients had a posterior-stabilised or cruciate-retaining Triathlon implant, performed with an image-based robotic-assisted system, with a resurfaced patella (Mako, Stryker Corp).

Data collection and variable selection

The assessed PROMs were knee injury and osteoarthritis outcome score for joint replacement (KOOS JR) Scores, Western Ontario and McMaster Universities osteoarthritis index (WOMAC) function and WOMAC pain preoperatively and at 1 year postoperatively. Data describing the patient's demographics (gender, age, height, weight, BMI and health insurance plan), diseased anatomy and the surgery were gathered as variables to predict RA-TKA outcomes (Figure 1).

TABLE 1 Mean and standard deviation for all input features.

Variable	Mean	SD
Numerical variables		
Age	67.9	7.9
BMI	32.0	6.8
Surgery duration (min)	65.6	12.8
tibia lateral resection depth (mm)	7.01	1.77
Femur posterior lateral resection depth (mm)	7.70	2.17
Femur distal lateral resection depth (mm)	5.76	2.47
Tibia medial resection depth (mm)	5.39	1.61
Femur distal medial resection depth (mm)	8.22	2.34
Femur posterior medial resection depth (mm)	9.44	2.48
Native mHKA (deg)	−3.49	5.41
Femoral translation anterior/posterior (mm)	3.86	2.75
Femoral translation distal/proximal (mm)	1.31	2.22
Femoral alignment internal/external (deg)	0.79	2.94
Tibial alignment varus/valgus (deg)	−0.49	1.35
Femoral alignment varus/valgus (deg)	−0.10	1.24
Tibial alignment internal/external (deg)	−0.11	1.30
Femoral alignment flexion/extension (deg)	−4.32	1.51
Femoral translation medial/lateral (mm)	2.01	2.16
Tibial translation medial/lateral (mm)	−0.21	2.21
Tibial translation anterior/posterior (mm)	−5.85	2.67
Tibial translation distal/proximal (mm)	10.42	2.31
Preop to postop HKA JLO Change (angle) (deg)	−4.44	32.13
Preop to postop HKA JLO Change (Distance) (mm)	7.91	3.28
B-Score	4.72	2.30
Tibiofemoral medial JSW	−2.55	2.93
Tibiofemoral lateral JSW	0.27	1.98
Patellofemoral medial JSW	−0.11	1.91
Patellofemoral lateral JSW	−2.27	1.84
Medial congruency	2.04	1.11
Lateral congruency	3.05	3.55
Trochlear ML Fit	3.42	1.13
Medial concentricity	3.70	2.38
Lateral concentricity	3.77	2.25
Categorical variables		
Gender	Female: <i>n</i> = 263 (62.2%)	
Laterality (left/right knee)	Right: <i>n</i> = 230 (54.4%) Left: <i>n</i> = 193 (45.6%)	

Variable	Mean	SD
Primary payer	Medicare: <i>n</i> = 169 (40.0%) Commercial: <i>n</i> = 157 (37.1%) Managed Medicare: <i>n</i> = 84 (19.9%) Medicaid: <i>n</i> = 7 (1.7%) Other: <i>n</i> = 6 (1.4%)	
Inpatient/outpatient setting	Inpatient: <i>n</i> = 364 (86.05%) Outpatient: <i>n</i> = 59 (13.95%)	
Femur implant type	CR: <i>n</i> = 423 (100.0%)	
Tibia implant type	Cruciform: <i>n</i> = 423 (100.0%)	

Abbreviations: BMI, body mass index; HKA, hip-knee-ankle angle; JLO, joint line obliquity; JSW, joint space widths; ML, mediolateral; mHKA, mechanical hip-knee-ankle angle.

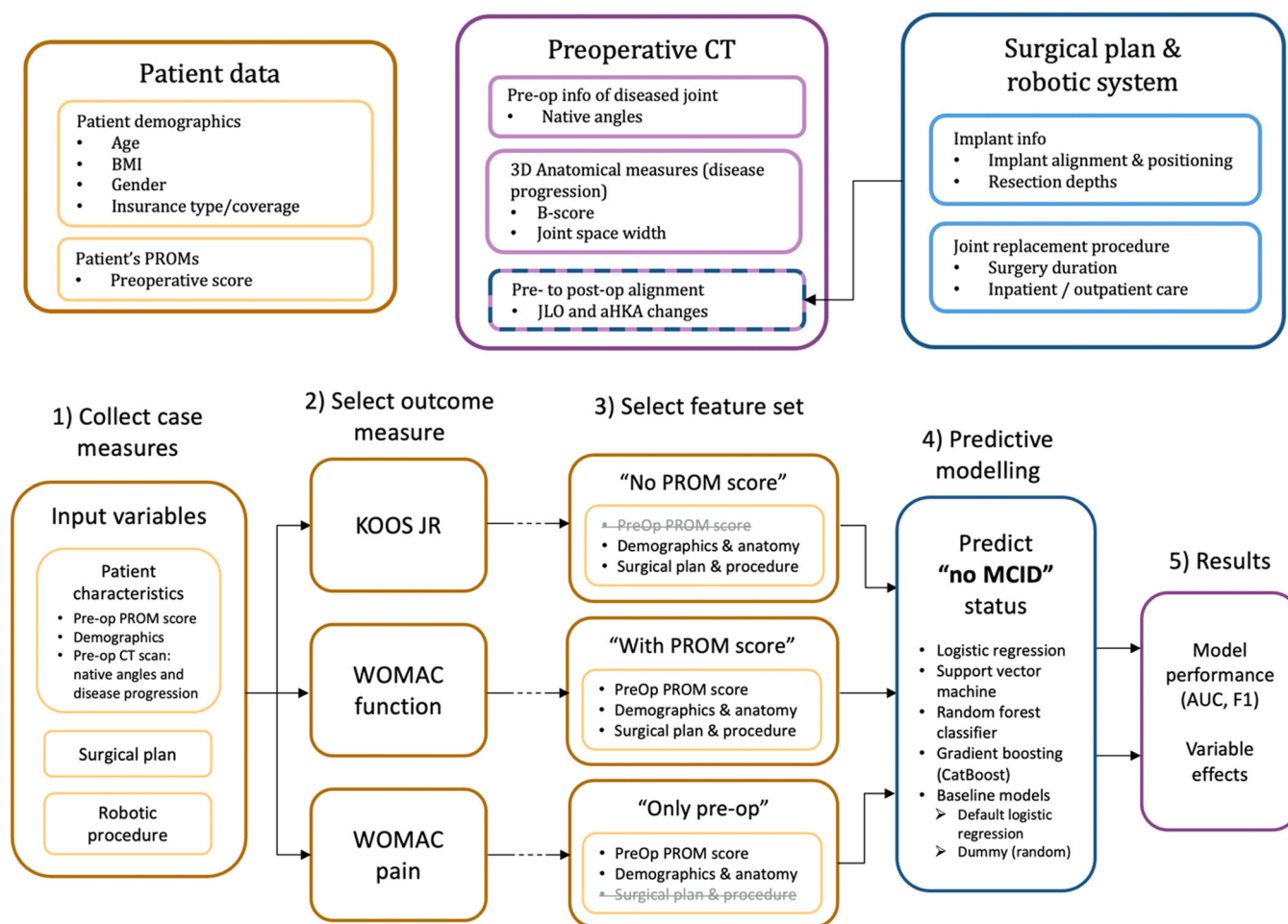


FIGURE 1 Predictive modelling setup to predict robotic-assisted total knee arthroplasty (RA-TKA) outcomes. The model predicts the target variable of not reaching a minimal clinically important difference (MCID), where MCID was calculated based on 1-year patient-reported outcome measure (PROM) score change in KOOS JR, WOMAC function and WOMAC pain. Input variables included measures related to patient demographics and the surgical procedure. Several machine learning models were trained for each PROM type and combination of input variables and were evaluated using the F1 score and area under the receiver operating characteristic curve (ROC-AUC) score.

The preoperative computed tomography (CT) scan was used to determine the patient's diseased anatomy. These include joint line obliquity (JLO), joint line congruency angle (JLCA), lateral distal femur angle

(LDFA), medial proximal tibial angle (MPTA) and arithmetic hip-knee-ankle angle (aHKA) and mechanical hip-knee-ankle angle (mHKA) [27]. These were also collected intraoperatively from the robotic system at the

postbalancing state, allowing calculation of the pre-operative to postoperative change in the JLO-aHKA plane [27]. Moreover, the following 3D anatomical features were calculated from the CT scan: (1) femur bone shape (B-score) to measure osteoarthritis progression [9] and (2) tibiofemoral and patellofemoral medial and lateral joint space widths (JSWs) to quantify cartilage loss [9, 10].

Variables characterising the surgery were inpatient/outpatient setting, surgery duration (time between incision and closure), ligament balancing duration, operation side, femoral and tibial component sizes and types (cruciate retaining (CR)/posterior stabilised (PS) and cruciform/universal) and polyethylene thickness. The alignment technique was personalised, similar to the current functional alignment described recently in the literature [26]. The final implant position and femoral and tibial resection depths were obtained from the robotic system. Lastly, features measuring 3D implant fit were calculated, including congruency (medial and lateral), concentricity (medial and lateral) and trochlear mediolateral (ML) fit (Figure 2). Congruency and trochlear ML fit were computed as the root mean square difference between the bone and implant surface along the medial and lateral condyles and trochlear dwell, respectively. Concentricity was calculated as the distance between the bone and implant knee centres.

Variables with high collinearity were removed to allow training and interpretability of linear models. The

following variables were excluded due to high correlations (correlation coefficient larger than 0.7): femoral and tibial component sizes, polyethylene thickness, patient height and weight, preoperative JLO, JLCA, LDFA, MPTA and aHKA and all postbalancing angles, ligament balancing duration and tibial slope. The final predictor variables and their distribution are listed in Table 1.

Outcome definition

The bicategorical target variable for the prediction model was defined as whether a patient's improvement in PROMs at 1-year post-TKA reached an MCID. Using an anchor-based approach, MCID thresholds were determined based on published studies with similar patient cohorts. An MCID of 15 was chosen for the WOMAC pain and function subscales and a value of 20.8 for KOOS JR [23]. A total of 79% of patients achieved an MCID on the WOMAC function scale and 75% on the WOMAC pain scale (Table 2). The MCID achievement rate for KOOS JR was 67%.

Predictive modelling

Classification models for each PROM were developed, with the aim of discriminating between patients who did not achieve MCID (target class) and patients who

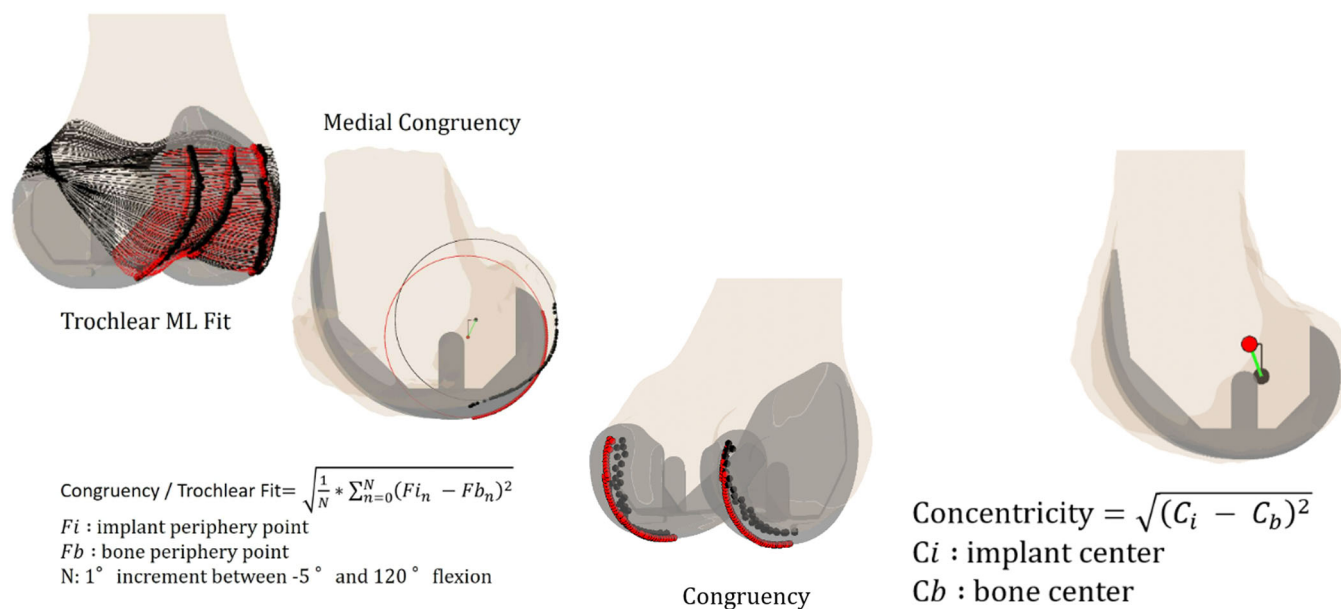


FIGURE 2 Visual representation of three-dimensional (3D) implant fit parameters included to describe implant-to-bone fit in 3D space. These variables were included to model the likelihood of not achieving a minimal clinically important difference (MCID) status 1 year after a robotic-assisted total knee arthroplasty (RA-TKA) procedure (Figure 1). Congruency represents the sagittal fit between the implant and native bone condyles computed from 10° to 110° of the flexion arc at 1° intervals. Trochlear mediolateral (ML) fit means the medial-lateral alignment between the native and implant trochlear dwell computed from -10° to 80° of the flexion arc at 1° intervals. Concentricity represents the fit of the bone and implant knee centres.

TABLE 2 Distribution of PROM scores and MCID status for the patient population used to model postoperative MCID status at 1 year after RA-TKA procedures.

Characteristics	KOOS JR	WOMAC function	WOMAC pain
# cases with pre-op and 1-year post-op score	385	344	344
Preop score (mean $\pm$ SD)	50.0 $\pm$ 12.0	55.1 $\pm$ 15.7	67.5 $\pm$ 10.6
1-year postop score (mean $\pm$ SD)	78.8 $\pm$ 15.6	85.9 $\pm$ 15.1	91.7 $\pm$ 10.6
1-year score change (mean $\pm$ SD)	28.8 $\pm$ 17.4	30.9 $\pm$ 18.7	24.2 $\pm$ 12.3
# cases reached MCID	257 (67%)	273 (79%)	259 (75%)
MCID threshold	20.8	15	15

Note: PROMs scores have been converted to a scale of (0, 100), where 0 represents the worst and 100 is the best possible score.

Abbreviations: MCID, minimal clinically important difference; PROM, patient-reported outcome measure; RA-TKA, robotic-assisted total knee arthroplasty.

achieved MCID at 1-year follow-up (Figure 1). The target class was set to 'not reaching an MCID' (class = 1). Three different feature sets were defined using the abovementioned variables as predictor variables. The first two variable sets included all preoperative, intraoperative and postoperative variables describing the patient demographics, diseased anatomy and the surgery. The preop PROM score was added to the set of predictor variables for the second variable set only [4]. The preop PROM score was excluded from the first variable set. Lastly, the third variable set consisted only of preoperatively available variables, meaning that implant- and surgery-related factors were excluded.

A variety of linear and nonlinear machine learning models were trained and evaluated for each combination of PROM and variable set. Machine learning algorithms were selected based on previous studies [24]. L2-penalised logistic regression was chosen for linear classification methods. Nonlinear models included support vector machine, random forest and CatBoost, which is a gradient-boosting algorithm designed to support categorical features. Two baseline methods were developed: (1) 'Dummy' classifier that randomly predicts MCID status based on the prior class distribution and (2) logistic regression without hyperparameter optimisation (default implementation from scikit-learn).

Model training and evaluation

The data were randomly split into a training data set (80% of the cases) and a test set (20% of the cases). This division was stratified based on the target class (MCID/no MCID) to ensure that the proportion of each class was maintained in both the training and the test set. For each machine learning model, a nested cross-validation scheme was used for training and evaluation; the inner-loop consisted of a 10-fold

cross-validation scheme for hyperparameter optimisation using a Bayesian approach (hyperopt package). The model with the best performing hyperparameters was chosen for the outer loop, which consisted of a threefold cross-validation scheme for performance evaluation. Finally, the test set was used in the final evaluation to give an unbiased estimate of model performance.

Model performance was evaluated by area under the receiver operating characteristic curve (ROC-AUC) and class discrimination (F1-score). Sensitivity and specificity were also calculated. 95% confidence intervals were calculated based on 5000 bootstrap resamples. The best performing models were further analysed by variable importance and variable-target relationships.

All analyses were performed using Python 3 (Python Software Foundation, <https://www.python.org/>, version 3.9.5). The hyperopt (version 0.2.7), scikit-learn (version 1.3.0) and catboost (version 1.2.1) packages were used for hyperparameter optimisation, model training and evaluation. Scipy.stats (version 1.7.3) package was used for statistical analysis, and plots were generated using seaborn (version 0.11.2) and matplotlib (version 3.5.1). Statistical significance was defined as $p < 0.05$.

Ethical approval

All procedures were performed in accordance with the ethical standards of the institutional and/or national research committee, the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. Informed consent was obtained for each patient in the study. Data collection and analysis were carried out in accordance with the authors' local institutional board. This study was approved by the South County Institutional Review Board on 29 May 2020.

RESULTS

Predictive models

Overall, the best performing model was L2-regularised logistic regression on the reduced WOMAC function PROM, using the full preoperative and intraoperative feature set. The model achieved an area under the curve (AUC) of 0.80 and an F1 score of 0.48 (Table 3). In general, adding the preoperative PROM score improved model performance (Figure 3). Models utilising the preop PROM score also outperformed the 'Dummy' baseline classifier.

KOOS JR and WOMAC function models showed stronger predictive performance when including intraoperative variables describing the surgical intervention, whereas only preoperative predictors yielded the best results for the WOMAC pain subscale.

Predictive variables

The top five variables most strongly contributing to not achieving MCID status were high preoperative PROMs scores, more valgus alignment of the femoral component, increased femoral posterolateral resection depth (related to internal rotation of the femoral component), high BMI and increased medial patellofemoral JSW (Figure 4). A larger variable coefficient means an increased likelihood that the patient will not reach an MCID. For instance, the odds ratio for the preop PROM score is 4.9, meaning if a patient scores one unit higher on the preop PROM, their odds of no MCID are

increased by 4.9. Therefore, the higher the preoperative PROMs, the greater the risk of not achieving the MCID status.

Conversely, the top five variables increasing the likelihood of achieving an MCID were an increase in medial shift of the tibial component, whether the primary payor was managed Medicare, high lateral congruency, high tibiofemoral medial JSW and increase in tibial medial resection depth, of which only tibial medial translation and managed Medicare insurance were statistically significant.

Less predictive variables included the patient's preoperative mHKA, trochlear ML fit, gender and surgery side.

DISCUSSION

The main findings of this study were that (1) preoperative PROMs and surgical data improved the predictive models of PROMs' improvement at 1-year post-TKA and (2) the main predictive factors to not achieving MCID status were high preoperative PROMs, increase in femoral posterolateral and distal-lateral resection depth and high BMI.

Similar results are seen in the literature for outcome prediction after TKA surgery (e.g., AUCs between 0.6 and 0.75 for prediction of MCID [KOOS JR]) (Table 3) [14, 16]. This matches the best KOOS JR model with an AUC of 0.75 in this study. The literature reported AUC scores of 0.8 and 0.89 for the WOMAC PROM [40]. A notable difference in methodology is that the analysis focused on identifying patients at risk of not reaching MCID status. Most studies in the literature

TABLE 3 Summary of the best performing models predicting MCID status at 1 year after RA-TKA procedures, per PROM type.

	WOMAC function	WOMAC pain	KOOS JR
Best model	Logistic regression, SVM	CatBoost	CatBoost
Feature set	With PROM score	With PROM score	With PROM score
AUC (95% CI)	0.80 (0.79–0.81)	0.79 (0.77–0.80)	0.75 (0.74–0.76)
F1 (95% CI), decision threshold = 0.5	0.48 (0.47–0.49)	0.44 (0.39–0.44)	0.62 (0.62–0.63)
Sensitivity (95% CI)	0.79 (0.77–0.81)	0.35 (0.30–0.35)	0.92 (0.92–0.93)
Specificity (95% CI)	0.62 (0.60–0.63)	0.92 (0.92–0.93)	0.47 (0.45–0.48)
Literature-reported performance	Hawker et al. [17]. AUC = 0.80 (primary total joint arthroplasty, i.e., hip and knee) Zhang et al. [40]. AUC = 0.88 using only preoperative WOMAC score. AUC = 0.89 with demographics and comorbidities as additional predictor variables.		Harris et al. [16]. AUC = 0.60 Fontana et al. [14]. AUC = 0.75
Comments	Literature reports accuracy in predicting 'MCID' class. This study reports 'no MCID' accuracy. MCID status in overall WOMAC score is predicted; therefore, there is no differentiation between pain and function subscales.		Literature reports accuracy in predicting 'MCID' class. We report 'no MCID' accuracy.

Abbreviations: AUC, area under the curve; MCID, minimal clinically important difference; PROM, patient-reported outcome measure; RA-TKA, robotic-assisted total knee arthroplasty.

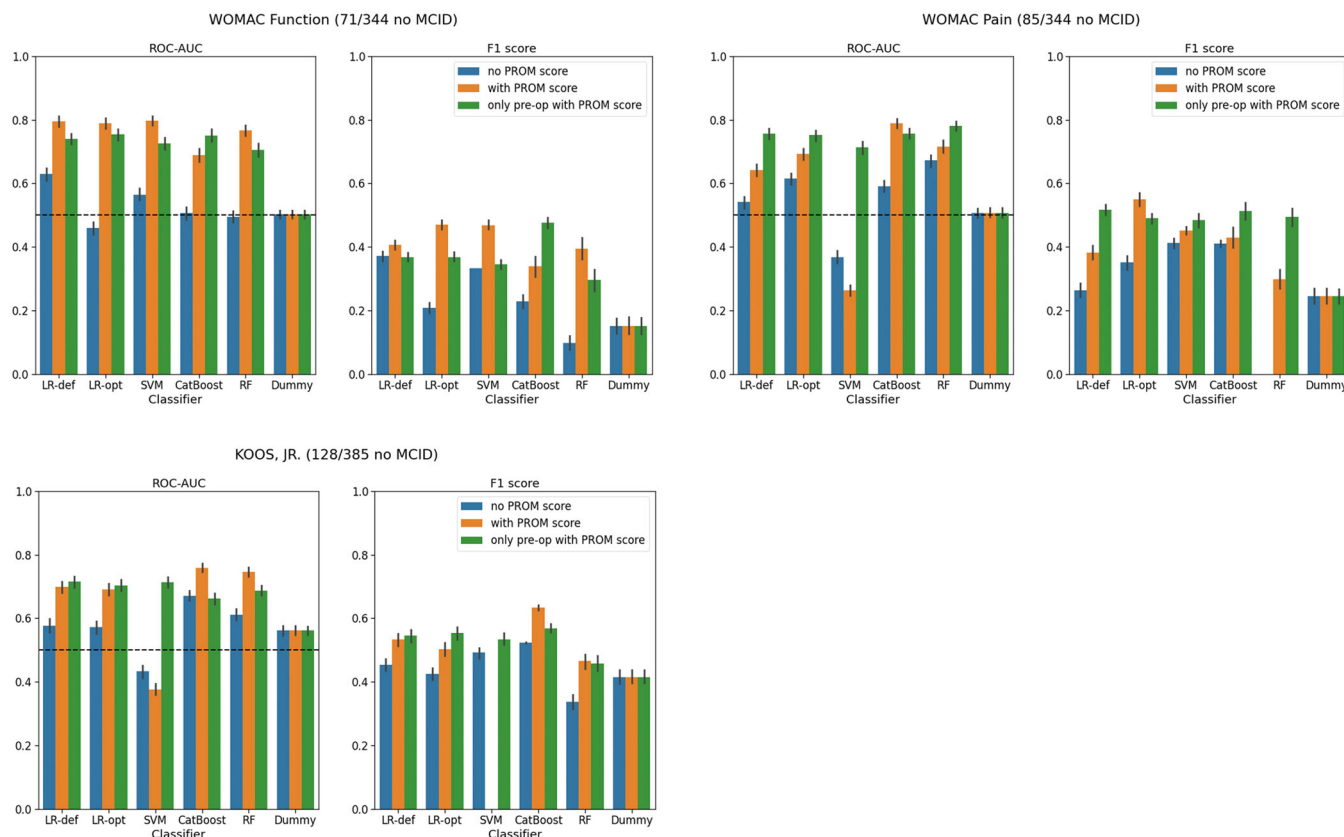


FIGURE 3 Machine learning models trained to predict nonminimal clinically important difference (MCID) outcomes for three different patient-reported outcome measures (PROMs) across three different variable sets. The three different PROMs were WOMAC function, WOMAC pain and KOOS JR. The three different variable sets were full preoperative and intraoperative variable set with preoperative PROM score, full variable set without PROM score and only preoperative variable. Model performance was measured using area under the receiver operating characteristic curve (ROC-AUC) and F1 scores on a holdout evaluation set and reports ability to predict 'no MCID' outcomes.

reported the MCID prediction accuracy. This difference between our methodology and the literature methodology can lead to performance score differences.

In the literature, most of the predictive models for TKA outcomes are based on patient demographics and preoperative data [4, 16, 19]. These studies show that preoperative PROMs have a strong and apparent association with the outcomes after TKA [2, 11, 34, 37, 39], as was also the case with the models used in this study. Including these PROMs improved predictive models' quality and performance. Distinct from existing studies, the models used in this study also included surgical planning and the disease-related variables, which further improved model performance, particularly for the prediction of the WOMAC function and pain scores. Predictive models have been developed in the literature on robotic-assisted TKA but without analysing the data extracted from the robotic system [13], except to predict the operative time of robotic-assisted TKA [28]. The advantages of the models used in this study were the inclusion of these patient's diseased anatomy and the surgical parameters. Nevertheless, the models developed in this study need some adjustments for clinical practice, particularly a validation in a

multicentric cohort or the data from other orthopaedic departments, and an easy application that can be used in daily practice. Some predictive models in the literature have been developed in applications or websites to facilitate their use in clinical practice [30, 36, 37]. However, most of them were neither intuitive nor feasible in clinical practice. In this study, the variables most strongly contributing to not achieving MCID status included three kinds of data: the preoperative PROMs, the BMI and the surgical planning (i.e., implant planning, bone resection plan and implant design). In the literature, several studies have described that the more severe the osteoarthritis and the lower the preoperative PROMs, the greater the improvement of PROMs and satisfaction after the surgery [1, 25, 37, 39]. This study confirmed that patients with high preoperative PROMs were more at risk of not achieving MCID in PROMs at 1 year postoperatively. These patients could be identified preoperatively as having a poor indication of TKA at this time or at least can be informed of the likely partial improvement of their symptoms.

BMI has also been assessed in the literature and was described as an inconsistent predictive variable.

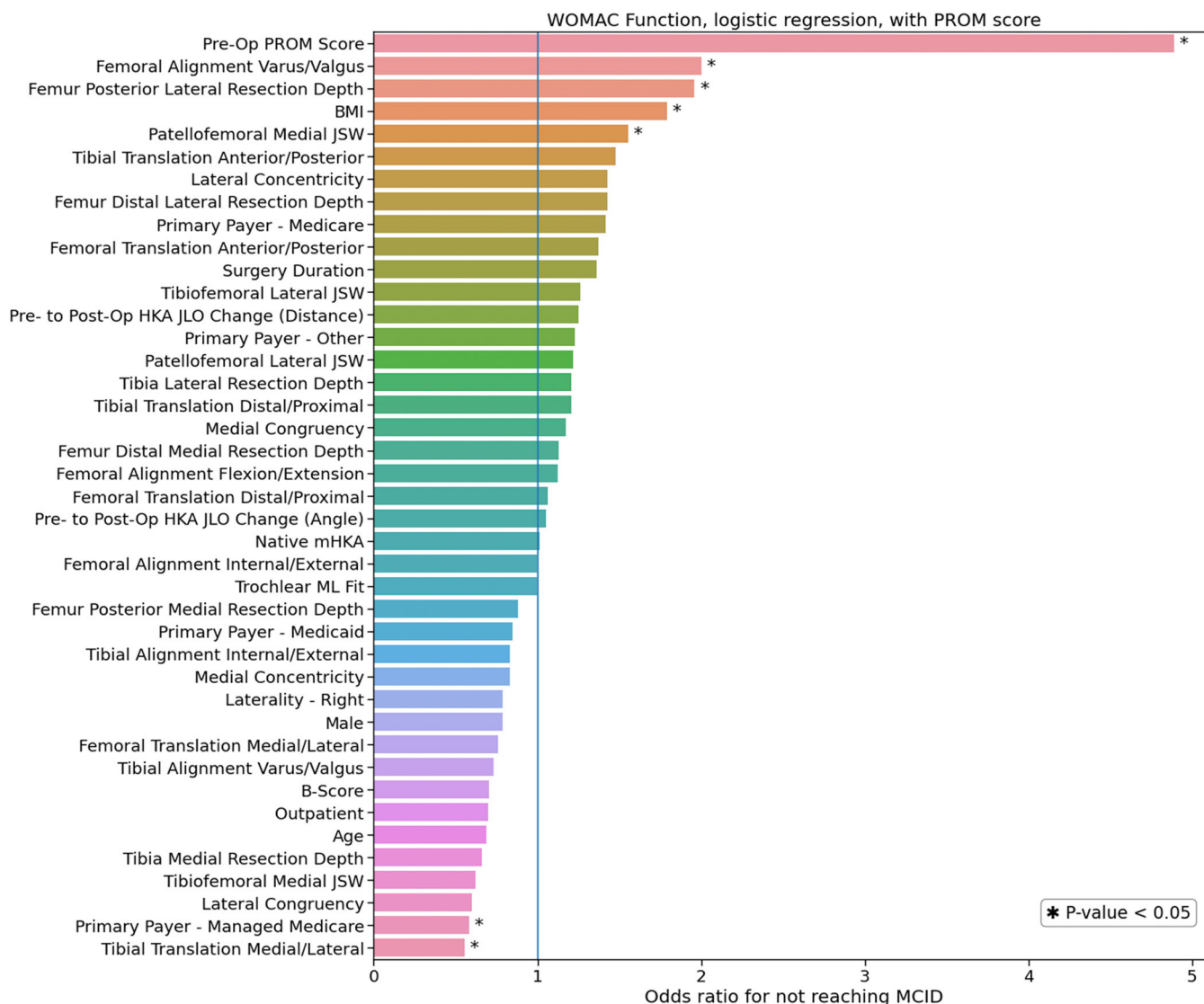


FIGURE 4 Logistic regression model trained to predict nonminimal clinically important difference (MCID) outcomes for the WOMAC function patient-reported outcome measures (PROMs). Adjusted odds ratios for the predictor variables were determined to analyse variable outcome relationships. The asterisk denotes that the variable coefficient is statistically significant ($p < 0.05$). The vertical line at odds ratio = 1 provides a reference for variables that have minimal effect on the target variable (no MCID outcome). An odds ratio > 1 indicates that the variable is associated with non-MCID outcome, meaning that a larger value results in a larger likelihood of not reaching an MCID. Odds ratio < 1 signifies that the feature reduces likelihood of non-MCID outcome, that is, it is associated with reaching an MCID.

Dowsey et al. described poorer functional outcomes in morbidly obese patients [12]. Nevertheless, this correlation is not consistently reported by all authors. In this study, the patients with a high BMI had a higher risk of not achieving MCID status. Contrary to some studies in the literature assessing only postoperative PROMs [35], this study evaluated whether or not MCID was achieved, considering the preoperative status. Thus, BMI control before the TKA surgery is essential and should be recommended to this population.

The surgical planning was not assessed in the predictive models from the literature. In this study, the two main surgical predictive variables were the valgus alignment of the femoral component and the femoral

posterolateral resection depth. MCID status had a higher risk of not being achieved if the femoral component was more in valgus positioning or if the femoral posterolateral resection depth was large, meaning more internal rotation in the femoral component. These two surgical positions are sometimes obtained with kinematic alignment. Indeed, in the literature, several studies have described internal rotation and valgus of the femoral component during strict kinematic alignment [20, 31, 33]. These two surgical positionings directly affect the trochlea positioning, leading to potential poor patellar tracking [22], unsatisfied functional outcomes [29] and a higher risk of revisions [21]. The correlation between medial patella-femoral joint space and outcomes can

also demonstrate the consequences of patellar tracking [22]. Indeed, an increase in the medial patella-femoral joint space can be due to lateral translation or tilt of the patella, due to poor trochlea restoration. The results of this study suggested the advantages of personalised surgery during TKA, such as with the functional positioning technique, using three-dimensional (3D) planning. Functional positioning allows the restoration of bone deformity to obtain a satisfying ligament balance in extension and flexion [32]. But this functional positioning also considers restoring the trochlear shape, thanks to the surgical planning performed on the CT scan with the image-based robotic-assisted system. This restoration of the trochlear shape can improve the patellar tracking after the surgery [32]. The compromise between the femoral rotation to restore good ligament balancing in flexion and to restore patellar tracking is primordial and specific to the functional alignment and the image-based robotic-assisted TKA. These surgical parameters necessarily affect the functional outcomes after TKA. Including these surgical factors in the predictive models appears thus crucial to avoid confounding factors in interpreting functional outcomes after TKA [4].

The relationship between medial tibial translation and functional outcomes is less obvious. Several explanations can be possible, particularly femorotibial congruency, the extensor mechanism alignment with the tibial tubercle position or the ideal positioning of a symmetrical tibial implant with potential impingement with the popliteus tendon. More investigations are necessary to explain this correlation.

Preoperative deformity (native mHKA) was not identified as a predictive factor in this study. The final alignment and implant positioning seemed more crucial to obtaining good functional outcomes than the preoperative deformity in a cohort with a majority of varus deformities. The lack of severe preoperative deformity could explain this absence of correlation. Moreover, the preoperative deformity was assessed on a preoperative CT scan and not on standing radiographs. It would be interesting to also include the preoperative deformity on a standing full long-leg radiograph.

This study has several limitations. First, the analysed data came from a small cohort of patients operated with a similar surgical technique. Using these predictive models, built and validated with data from one restricted population in another population, can be inaccurate and irrelevant. This is why predictive models using data from large populations numbering in the hundreds of thousands, including several centres or countries, are more relevant and reliable. A second limitation is the number of variables considered to formulate the predictive model. This is a limitation for any machine learning model, where certain variables, possibly predictive of outcomes, were not included and thus not assessed. Nevertheless, this study has analysed new factors, such as surgical

planning, to improve the relevance of the predictive models. Third, the database used for this study did not include patients with a BMI superior to 40 kg/m², which is considered 'unusual'. This exclusion can decrease the impact of the BMI. However, there are few patients concerned by this criterion. Then, the relevance of the models of this study for outcome prediction after TKA surgery was not very high, although similar to the literature (with an AUC of 0.80 and F1 score of 0.48). However, they can still be improved by selecting or adding strong predictive factors, such as mental health, and a larger cohort of patients. For example, anxiety and depression before surgery are frequently identified as risk factors for lower patient-reported outcomes after TKA. Progressively, it is crucial to determine the strong predictive factors and adjust the predictive models.

CONCLUSION

The machine learning models to predict MCID achievement 1 year after TKA developed in this study demonstrated that the most predictive variables included preoperative PROMs, BMI and surgical planning. The surgical predictive variables were valgus femoral alignment and femoral rotation, reflecting the benefits of personalised surgery and the interest in including surgical factors in predictive models for functional outcomes after TKA to guide surgical decision-making for every patient.

AUTHOR CONTRIBUTIONS

Lisa Spahn Lundgren: Study design; analyses; development of prediction model; and manuscript writing. **Nathalie Willems:** Analyses; development of prediction model; and manuscript editing. **Robert C. Marchand:** Literature review and manuscript editing. **Cécile Batailler:** Study design; literature review; and manuscript writing. **Sébastien Lustig:** Study design; supervision; and manuscript editing. All authors read and approved the final manuscript.

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CONFLICTS OF INTEREST STATEMENT

Lisa Spahn Lundgren is an employee for Stryker. Nathalie Willems is an employee for Stryker. Robert C. Marchand is a consultant, sponsored researcher, institutional support receiver and member of the Surgeon Advisory Board with stock options from Stryker. Sébastien Lustig is a consultant for Stryker, Smith Nephew, Heraeus, and Depuy Synthes; He received Institutional research support from Groupe Lepine and Amplitude and is a member of the Editorial Board for Journal of Bone and Joint Surgery (Am). The remaining author declares no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are not openly available.

ETHICS STATEMENT

All procedures were performed in accordance with the ethical standards of the institutional and/or national research committee, the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. Informed consent was obtained for each patient in the study. Data collection and analysis were carried out in accordance with the authors' local institutional board. This study was approved by the South County Institutional Review Board on May 29, 2020.

ORCID

Cécile Batailler  <http://orcid.org/0000-0003-3990-0906>

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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